Navigating AI in evidence synthesis

AI tools across the review workflow

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Map and presentation

lukebax.github.io/ai-review-tools/

Stages of a systematic review

1 Protocol

2 Search

3 Screening for relevance

4 Quality assessment

5 Data extraction

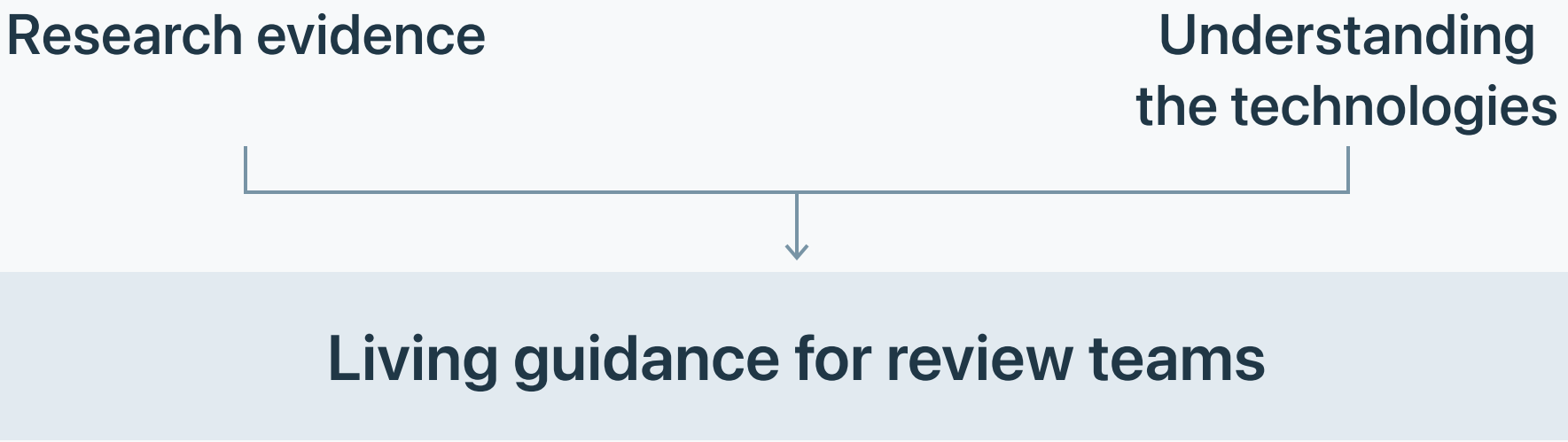
6 Data synthesis

7 Certainty of evidence

8 Manuscript

RAISE provides guidance for using AI in reviews

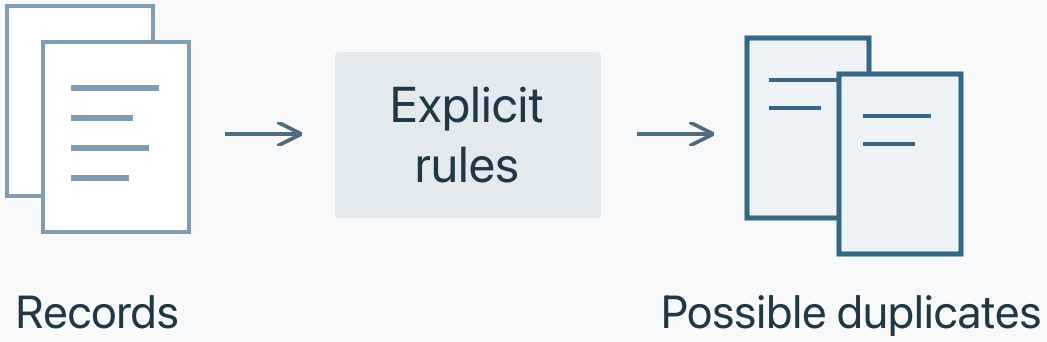
Responsible use of AI in evidence synthesis



Developed by researchers working in evidence synthesis

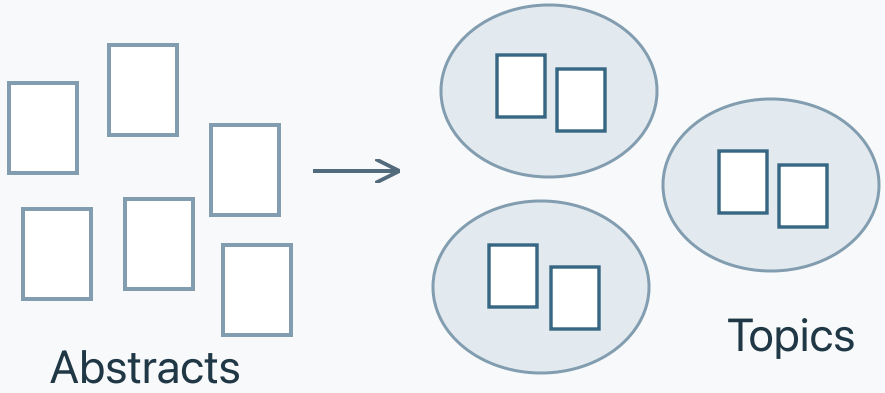
Explicit rules and patterns in unlabelled data

Rule-based algorithms



People define the rules used to compare records.

Unsupervised machine learning

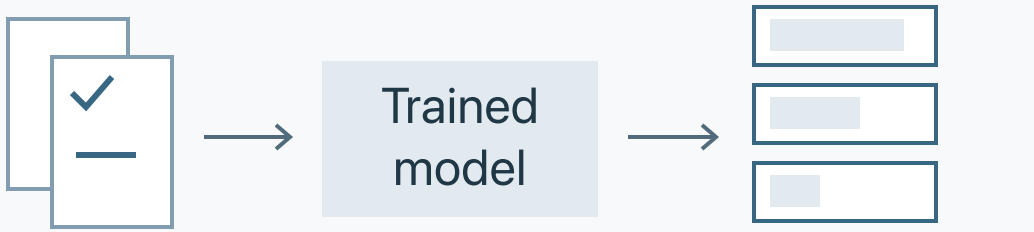


Patterns emerge without supplied labels.

Results depend on the records and the method.

Learning from labelled examples and generating language

Supervised machine learning

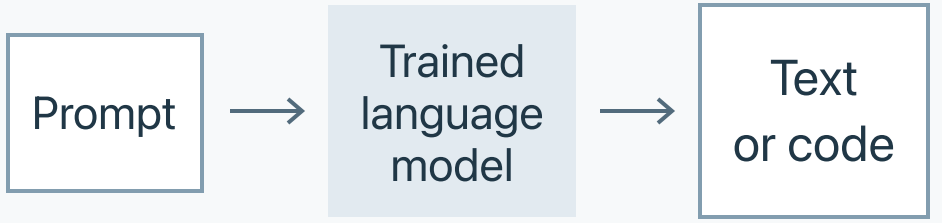


Labels

Relevance ranking

Already trained or trained
during the review

Generative LLMs

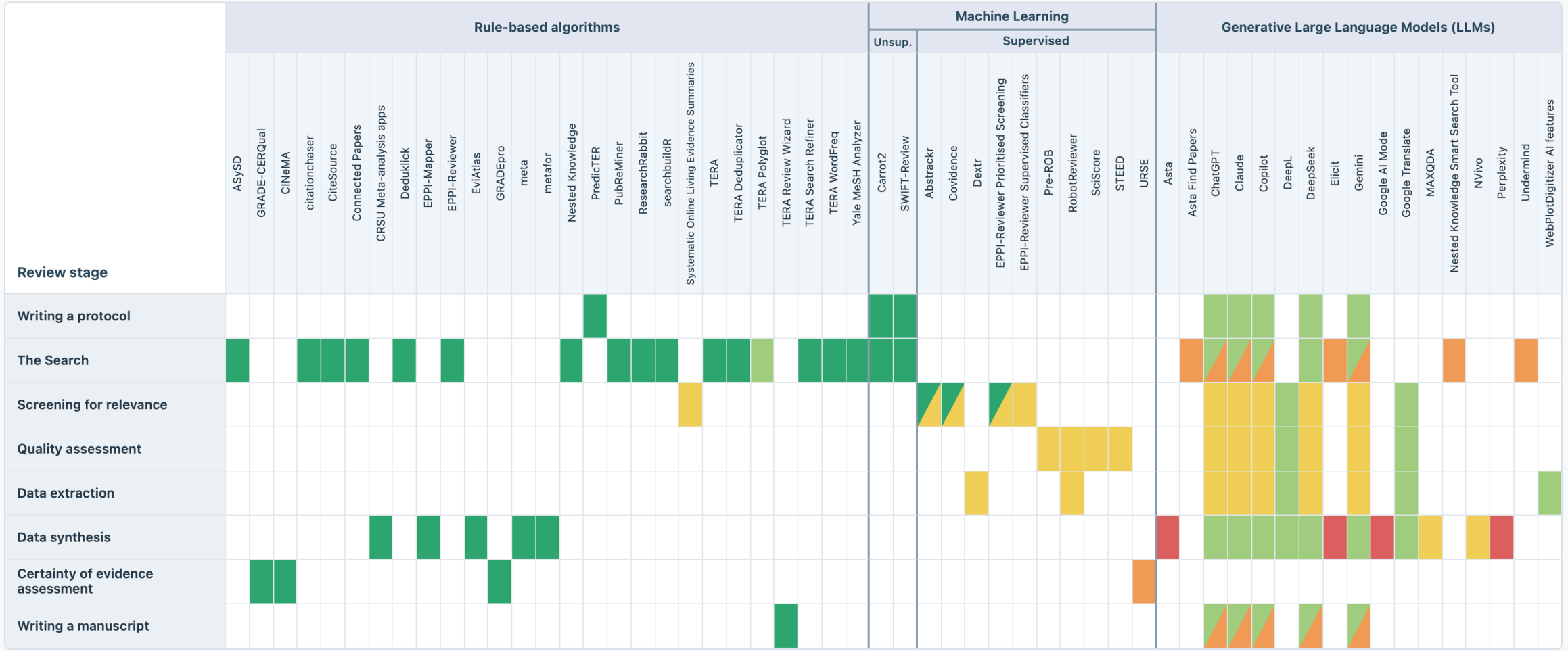


Outputs can vary.
Fluent output can be wrong.

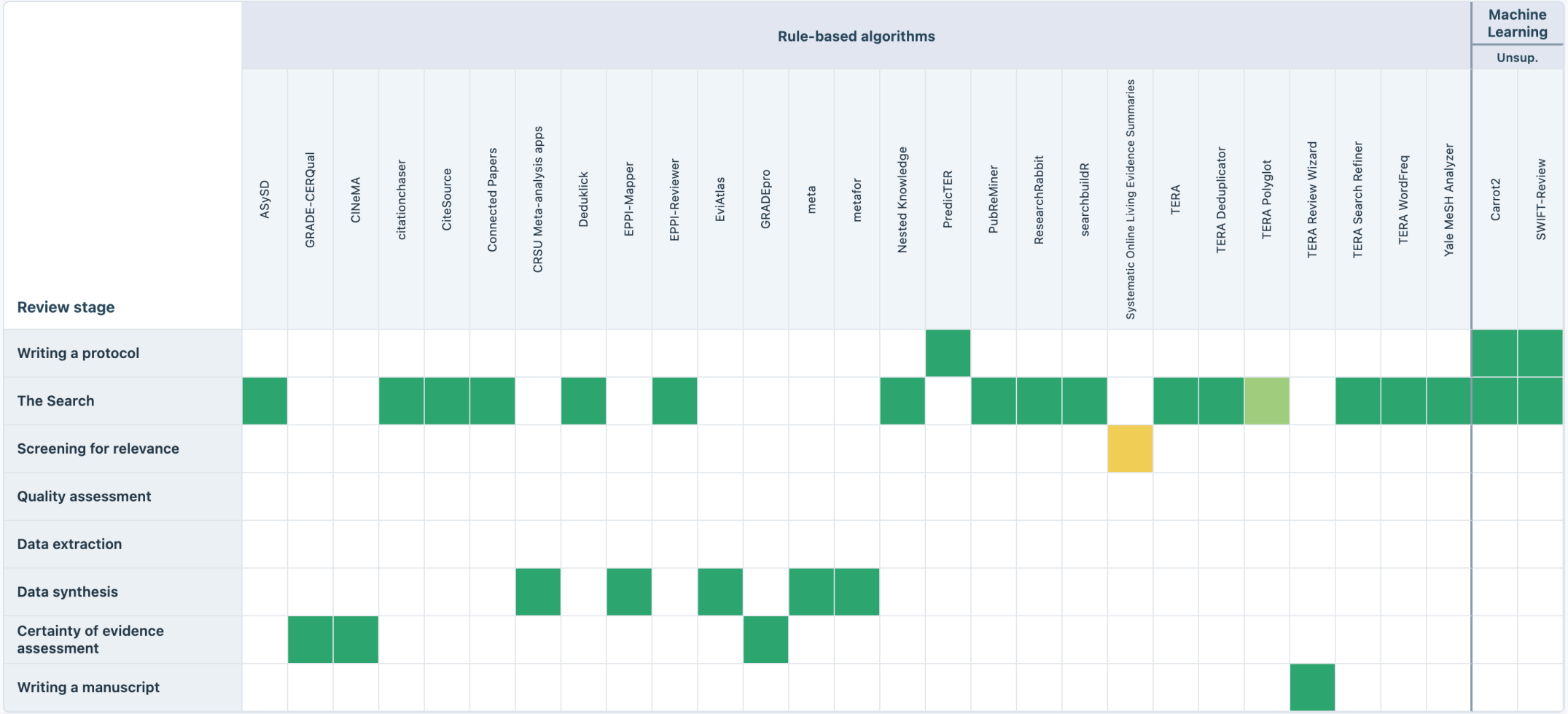
Recommendations apply to a particular use

Acceptable for use	Use outputs, accounting for limitations and bias
Human verification required	Check outputs and correct errors
Requires validation within the review	Test performance against a prespecified threshold and suitable benchmark
Exploratory and supplementary use	Explore or supplement established methods
Not acceptable for use	Do not rely on outputs for this task

AI tools across the review stages



Most of these mapped uses are acceptable




Machine Learning
Supervised
Review stage
Abstrackr
Covidence
Dextr
EPPI-Reviewer Prioritised Screening
EPPI-Reviewer Supervised Classifiers
Pre-ROB
RobotReviewer
SciScore
STEED
URSE
Writing a protocol
The Search
Screening for relevance
Quality assessment
Data extraction
Data synthesis
Certainty of evidence assessment
Writing a manuscript
RAISE use recommendations
Select one or more to highlight matching cells
Acceptable for use
Human verification required
Requires validation within the review
Exploratory and supplementary use
Not acceptable for use

Stopping early changes reliance on the ranking

Abstrackr



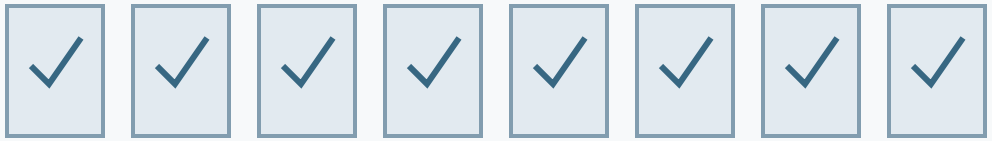
Covidence



EPPI-Reviewer Prioritised Screening

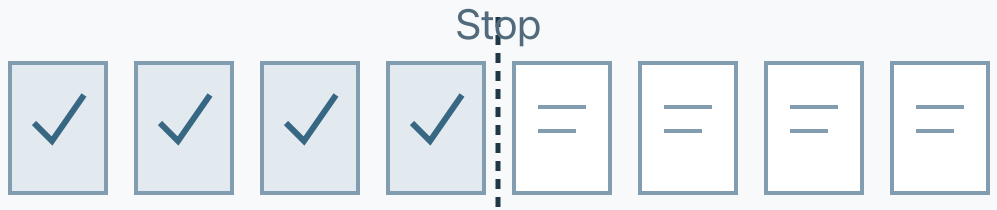


Screen every record



Every record is screened

Stop early



Screened

Unscreened

Acceptable for use

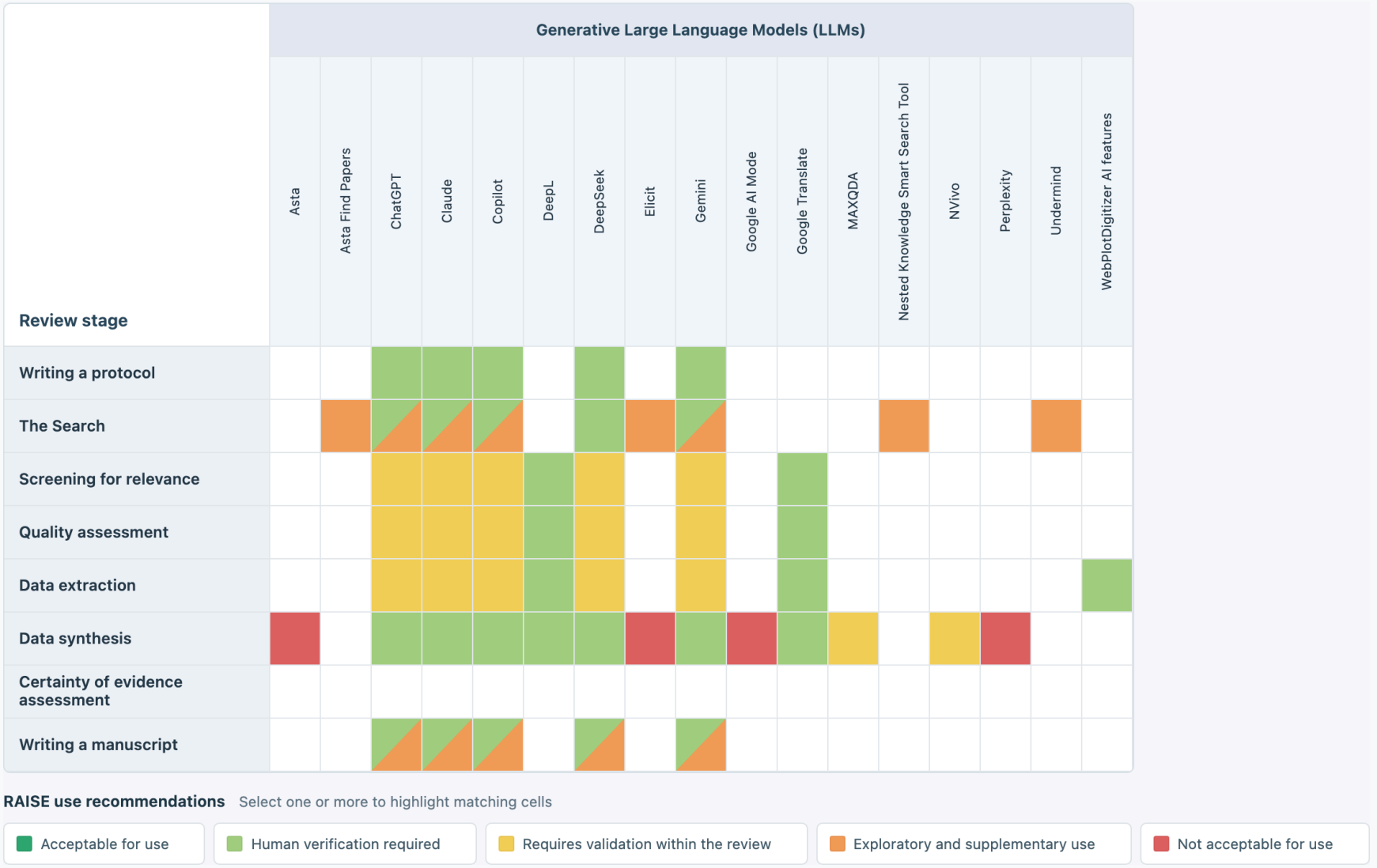
A poor ranking does not leave records unseen.

Requires validation within the review

Assess the screening and stopping procedure.

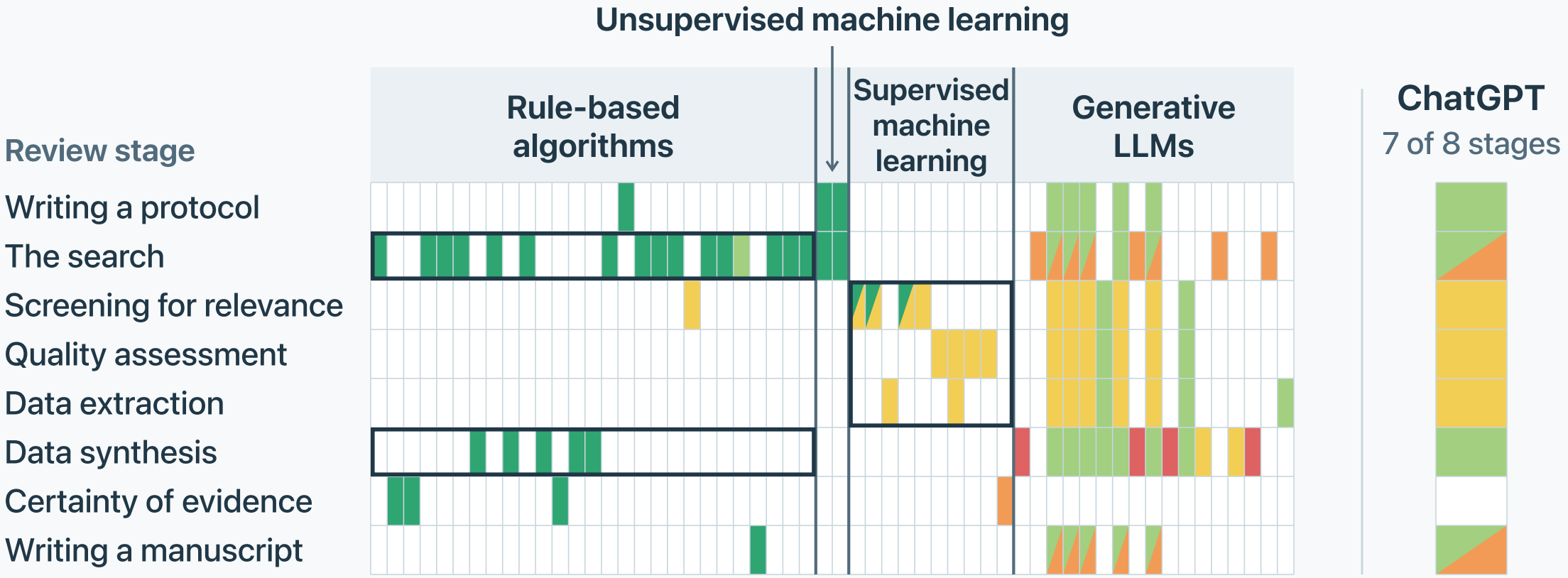
The ranking can be identical. What changes is how much we rely on it.

The mapped LLM uses carry conditions or restrictions



Tools differ in where and how broadly they are used

Stage clusters in the overview, with ChatGPT enlarged alongside



Rule-based
Search and synthesis

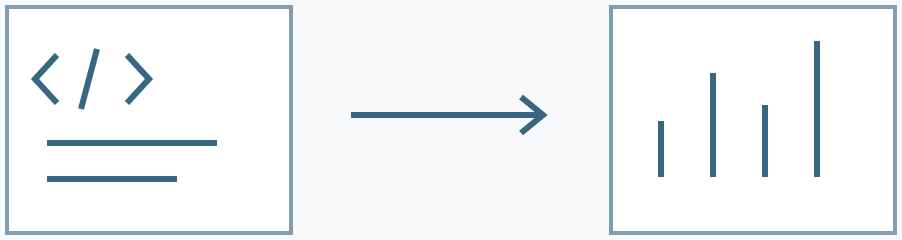
Supervised
Screening, quality assessment, extraction

ChatGPT
Mapped in 7 of 8 stages

Verification and validation ask different questions

Verification

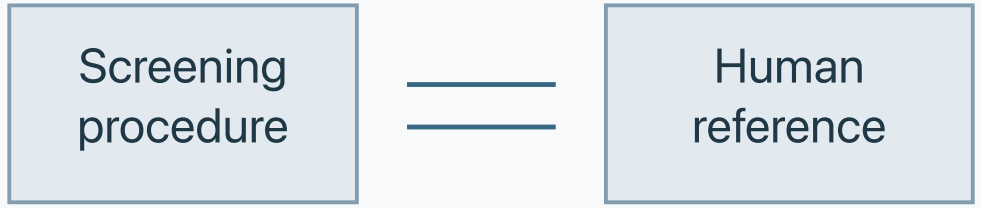
Check the output



Does it carry out the intended analysis?

Validation

Evaluate performance



Recall: proportion of eligible records found

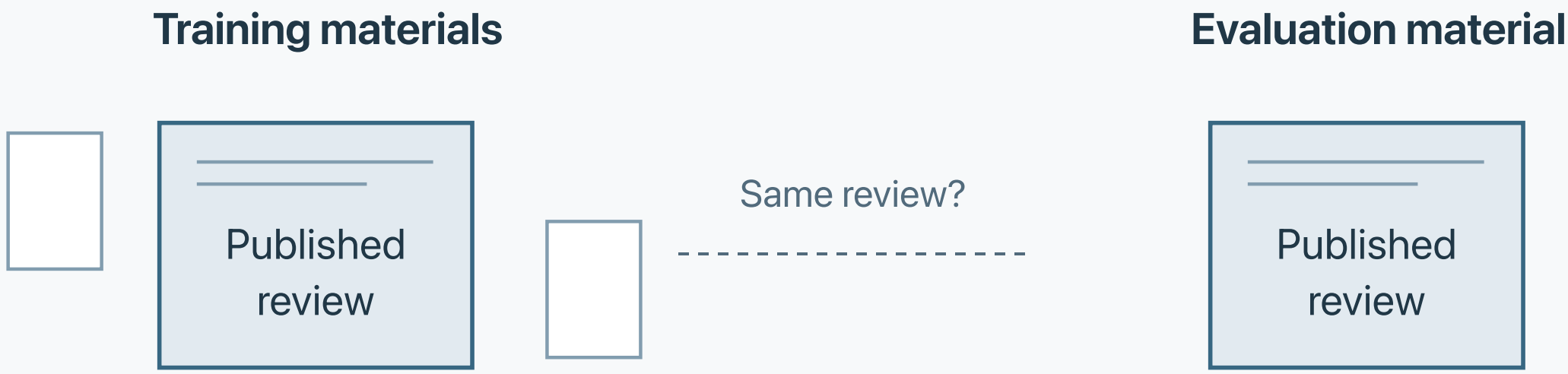
Applicable prior evidence may suffice. Follow any within-review requirement.

Suitable sampling or stopping methods can help, but assessment has a workload cost.

A workflow may need both.

Evaluation needs independence from development

Possible training–evaluation overlap: a hypothetical example



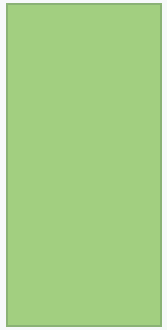
Apparent performance may overstate performance on new material.

Incomplete disclosure makes overlap harder to establish.

Two different uses during screening

Screening for relevance

Google Translate

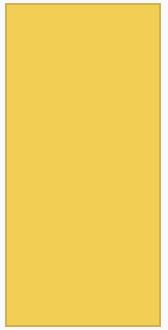


Translation support

Human verification required

Check the translation against the original.

Google’s Gemini



Relevance judgements

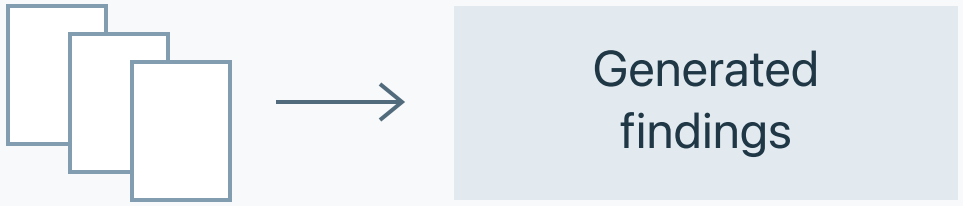
Requires validation within the review

Evaluate the screening procedure’s performance.

Selected screening cells from the map.

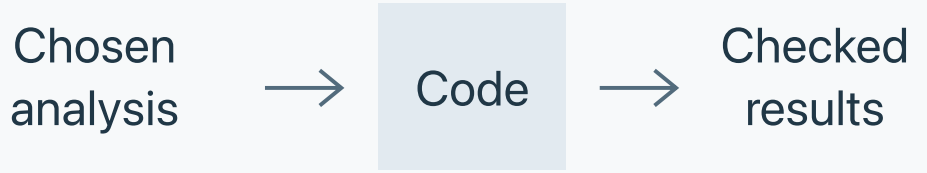
Synthesising findings differs from drafting analysis code

LLM-derived synthesis across studies



Not acceptable for use in a formal review

LLM-drafted code for a chosen analysis

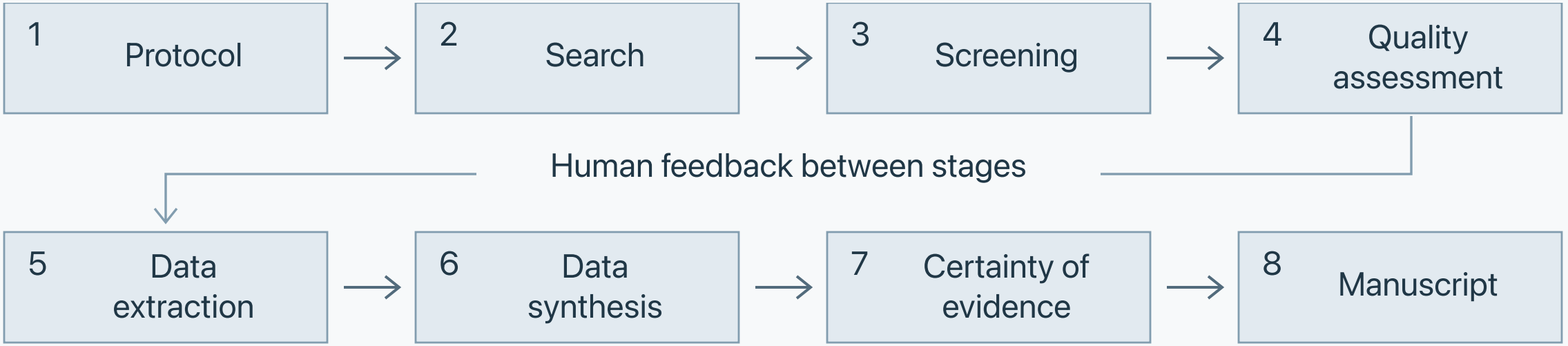


Human verification required

Qualitative coding or themes: validation and verification.
The boundary with delegated synthesis is not fully defined.

RAISE also restricts complete agentic reviews

Complete agentic review: not acceptable for use



Human feedback does not remove this restriction.

Individual tasks need separate assessment.

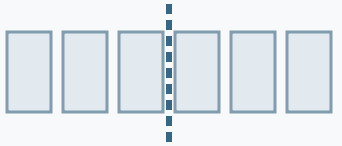
Is this tool suitable for my review?

Prioritised screening with early stopping

1

What is the purpose of the tool?

Assess ranking plus the stopping procedure.



2

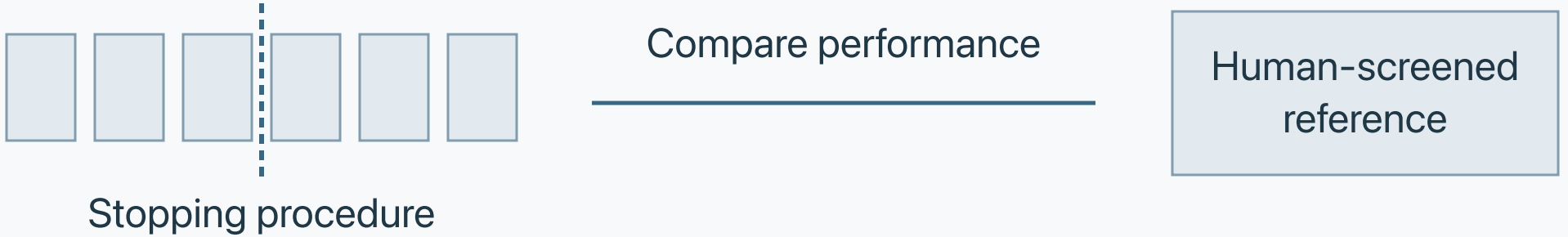
Where have the training, testing and validation data come from?

Review-team labels: quality and selection.
Method evaluations: relevance to this review.



3 Is the tool validated and performing sufficiently well for use?

Prioritised screening with early stopping



Evidence for the stopping procedure

Set required performance in advance.
Establish whether it is met within this review.

An appropriate human-screened reference

Team, resources and documentation

Prioritised screening with early stopping

4 Does the team have the expertise and resource to use the AI tool effectively, and is the usability acceptable?

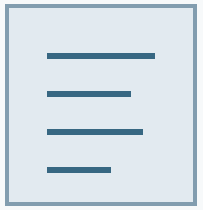
Skills, assessment time and a workable process

5 Is there sufficient transparency around licenses and other available documentation?

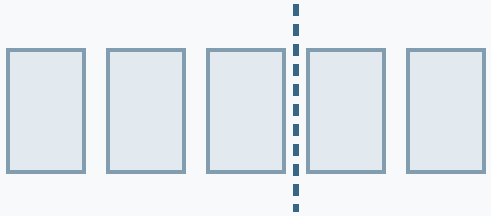
Instructions, access, data handling, version and settings

Missing stopping evidence must be resolved before reliance.

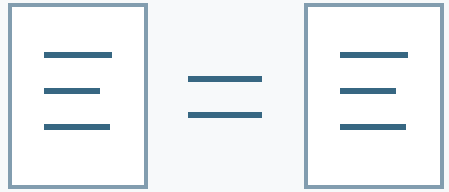
Choosing a suitable use of AI



Tool and task



Reliance on outputs



Evidence and checks

Map and presentation

lukebax.github.io/ai-review-tools/